Hamza Ghojaria

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Data Science experience candidate with **6+ years** of IT experience and a strong foundation in **Generative AI, ML, and Data Science.** Proven expertise in developing scalable **data-driven solutions** across finance, e-commerce, and cloud platforms using **Python, FastAPI, Azure, and SQL**.

Skills:

Data Science: Python, LLM(Azure OpenAI, Ollama), Generative AI, Machine Learning, SQL, Fast API, Analytics, Deep Learning, NLP.
Software: Visual Studio Code, Excel, Jupyter Notebook, Anaconda, Spyder, MSSQL, Tableau, Power BI.
Cloud: Azure (Containers, Blob, Document Intelligence, OpenAI)

Professional Experience:

LTIMindtree - Data Scientist: (Mar 2024 - July 2025)

- The Python-based document comparison application, built for a finance client, detects semantically similar sentences across various financial documents, ensuring accuracy and consistency in reports.
- **Document Compare Engine- FastAPI, Python, Pandas, Azure OpenAI, Azure Blob**
  The application extracts content from user-uploaded documents and **finds semantically similar sentences across different documents**. The engine lets users compare source and query files across formats like PDF, Word, JSON, and Excel. Input file formats, such as JSON and Excel, undergo pre-processing to create a common format. Also, other input file formats like PDF and Word documents are handled using **Azure AI Document Intelligence**, which converts them into a hierarchical JSON structure along with their metadata.. Based on the user's selection, the processed data is filtered, chunked, embedded, and stored in a Vector Database. Prompts are used to generate the retrieval process using Azure Search. The entire process is managed using multiple Api built on **FastAPI** for **the backend** and ReactJS for the **front end**. Errors were reduced to 80%, resulting in nearly a 70% reduction in time spent on traditional document processing.
- **PDF Extractor- FastAPI, Python, Pandas, Azure OpenAI, Azure Blob**
  The application extracts content from user-uploaded **PDF** documents and structures it in a predefined Excel format. Using the Aspose.PDF library, it extracts text along with the **metadata** (Text, Color, Font size, Bounding Box, etc). The extracted content is chunked based on the page. Each chunk is passed to **LLM (Azure Open AI GPT4o model)** along with asynchronous processing using **Asynico**. The **LLM** responses are structured using the Pydantic **model**. Also, the token usage is tracked and stored throughout the process.

Hansa Cequity - Associate Data Science and AI: (May 2022 - Mar 2024)

- Implemented a recommendation engine using Python for an e-commerce website by utilizing machine learning techniques to provide personalized product recommendations.
- Created propensity-based machine learning models (**Logistic Regression, Random Forest, XGBoost, Decision Tree**) to analyze large datasets and provide actionable insights for retail and e-commerce clients.
- Developed transaction data analysis tools to assess impact analysis and simulate business performance, helping clients optimize and track revenue and minimize expenditure.
- **Recommendation Engine-** Designed a recommendation engine to increase user retention on the website. The two models that were implemented were **popularity** and **sequence models**. As per the business problem, the sequence-based model is implemented based on the **FP growth algorithm**. For the popularity-based model, the popular product is based on the **buying frequency** in each Pincode /State /District. This resulted in a **50% increase** in user retention time and a **10% boost** in product sales.
- **Referral Model:** Designed and implemented a propensity-based machine learning model using the **XGBoost Algorithm**. It is capable of accurately predicting conversion likelihood for each prospect. The model contributed to a **25% increase** in the **referral rate** within a short timeframe.

CRISIL - Data Science Intern: (Dec 2021 - Apr 2022)

- Developed an end-to-end machine learning pipeline incorporating multiple Machine learning models.
- Implemented **Flask** using **REST API** to interact with the client-side application and present results.
- Implemented scalable architecture using **Python** for an **Accounting firm** based on keywords like Debt, Equity, Revenue, Cost, etc., to be **scraped** from the required documents and keep a check on optimal results.
- Worked on a business problem where a table needed to be **scraped** from the PDF's set of pages and used **LabelImg** to annotate those pages & pass the optimized outcomes for further development of the product.
- Designed and coded in Python to solve business problems, **data manipulation** & **data aggregation** for clients and teams.
- **Multiple ML Model Implementation-** Developed an end-to-end machine learning pipeline to conduct a comparative analysis of multiple machine learning model implementations simultaneously. Users can select a dataset where missing values are handled and followed by data visualization using the **Pandas Profiling library**. The pipeline lists many models from **Regression, Classification, and Clustering**. Upon selecting a category, all models are executed concurrently, providing accuracy metrics for each. The backend is built using **Python** and **Flask**, while the frontend is developed with **HTML, CSS, and JavaScript**. This

approach greatly reduces the team's efforts by **60%** when selecting the optimal machine learning model that delivers the **best results**.

Media.Net - Associate Software Test Engineer: (Mar 2019 - Mar 2021)

- Proficient in designing and maintaining **Python** scripts to automate test cases and template codes, leading to increased efficiency and reduced manual effort in testing and development.
- Implementing **Web Scraping** methods to extract desired data and monitor the parameters implemented by the development team.
- Demonstrated expertise in Python and its associated libraries and frameworks, resulting in improved product quality and reduced time-to-market for clients.

AlphaCentrix - Team Member QA & Support: (Jun 2018 - Feb 2019)

- Maintained and optimized **ETL tool using VBScript**, ensuring smooth operations through regular updates and code modifications.
- Executed data migration processes for the updated ETL tool, ensuring accurate data transfer and minimal client downtime.
- Managed multiple finance clients' operations, loading trades, equity, and bonds to the system while maintaining **SQL query updates** based on client requests.

Education:

- **Mumbai University: Sardar Patel Institute Of Technology (May 2020 - May 2022)**
  Master's of Technology in Computer Engineering ***(CGPA 8.1 / 10)***
- **Mumbai University: Rizvi College of Engineering (May 2015 - May 2018)**
  Bachelor's of Engineering in Electronics and Telecommunications ***(CGPA 6.1 / 10)***

Certifications:

Introduction to Generative AI Course: Google, Google Cloud Certified Professional Cloud Architect Course: Qwiklabs, IBM WatsonX Foundations Course: IBM, Introduction to Large Language Models Course: Google, Develop Generative AI solutions with Azure OpenAI Service: Microsoft, Text Prompt Engineering Techniques: Google, LangChain for LLM Application Development: Deep Learning.ai.